

Abstract of Scientific Justification (300 words max) (2 points): We want to measure the light curve of 1-2 supernovae, using the Mont4k CCD at the 61" Kuiper telescope, in the B, V, and (maybe) R filters. Light curves can tell us about the type of supernova and how it evolved over time, which helps us to determine certain characteristics of the star, like radius and mass. Any supernovae discovered recently will have faded by the time we plan to observe them, so we will select suitable targets from the Transient Name Server shortly before our observation period. Suitable targets should have a V mag less than 19, a declination between 40 and -5 degrees, and enough separation from the moon to avoid interference with our measurements. We will construct light curves with our data, and by comparing with light curves from other supernovae, we can measure peak brightness, color evolution, and the decay of radioactive elements produced in the explosion.

Telescope Requested (0.5 points): 61" Kuiper

Instrument (0.5 points): Mont4k CCD (B,V filters, R if possible)

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): We would prefer a dark or gray moon, since we are observing dim objects (18-19 mag). A bright moon could work if the supernova is especially bright and adequately separated from the moon, but it would reduce our SNR, so we would need to expose for longer to compensate.

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

Object Name	Right Ascension (h, m, s)	Declination (d, m, s)	V magnitude*	Object Type	Visibility
AT 2026aawd	20:21:37.093	+23:41:09.61	18.01	Transient (SN candidate)	September, October
AT 2026aars	02:29:34.883	+23:06:58.44	18:42	Transient (SN candidate)	September, October

* These are the discovery magnitudes, from the Transient Name Server.

These will probably not be the final targets we observe, since they would have faded by the time we actually go to the observatory, but these are examples of suitable targets for our project. Before our observation period (< 1 week) we will check the TNS database again for more recent candidates.

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? **(2 points)**

We want a SNR of at least 100 in each B, V, and R measurement. To calculate the exposure time, we can use $SNR \sim \sqrt{Ft}$, and take a ratio with a reference observation (subscript *our* denotes quantities for our target, subscript *ref* denotes quantities for a reference object):

$$\frac{SNR_{our}}{SNR_{ref}} = \sqrt{\frac{F_{our}t_{our}}{F_{ref}t_{ref}}}$$
$$t_{our} = \frac{F_{ref}t_{ref}}{F_{our}} \left(\frac{SNR_{our}}{SNR_{ref}}\right)^2$$

To make this useful, we can use the magnitude equation to get the ratio of fluxes in terms of magnitudes:

$$m_{ref} - m_{our} = -2.5 \log_{10} \left(\frac{F_{ref}}{F_{our}} \right)$$
$$\frac{F_{ref}}{F_{our}} = 10^{-(m_{ref} - m_{our})/2.5}$$

A past study (Xu et al. 2018) used the Mont4k instrument to obtain a SNR of 300 using a 30 second exposure on a 16.2 mag target. Using the fainter end of our usable targets, 19 mag, we can find our exposure time:

$$t_{our} = 30s \times 10^{(19-16.2)/2.5} \times \left(\frac{100}{300}\right)^2 \approx 45s$$

Since they used a white filter and we are using colored filters, we can guess that due to collecting fewer photons, a safe estimate of our exposure time is triple this, so around 135 seconds. To be extra safe, we will use a standard 240 seconds of exposure for each filter. We can refine this as needed at the observatory using reference stars.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? **(2 points)**

We will collect brightness measurements in each band, from which we can extract color and overall brightness. We can combine this with archival data or data from other groups with extra time in order to construct a light curve over time. The more data, the better, as we can more accurately plot the shape of the light curve. From this we can determine the rise and decline rates, and with good enough data, determine the type of supernova.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

If necessary, we can construct a light curve from archival and public data, from telescopes like ZTF. This would entail processing the data, making the light curve, and classifying it using photometric classification packages.

Reference:

Xu 许, S. 偲艺, Su, K. Y. L., Rogers, L. K., et al. 2018, ApJ, 866, 108,

<https://iopscience.iop.org/article/10.3847/1538-4357/aadcfe>

Graphs:

These graphs were made using Astroplan and the Jupyter Notebook template from the class GitHub. Using CSV data from TNS, we picked out some targets using our criteria and constructed some graphs showing altitude and airmass with respect to MST. The below graphs are for the two example targets outlined in the table previously.

